```
291
292  -- 1
293  CREATE INDEX idx_flights_act_departure ON flights(act_departure_time);
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 70 msec.
```

Query returned successfully in 70

Total rows: | Query complete 00:00:00.070 | CRLF

```
295
296  --2
297  CREATE UNIQUE INDEX idx_flights_no_sch_dep ON flights(flight_no, sch_departure_time);
298
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 42 msec.
```

```
299
300  --3
301  CREATE INDEX idx_flights_der_app ON flights(departing_airport_id, arriving_airport_id);
302
303
```

Data Output | Messages | Notifications

```
CREATE INDEX

Query returned successfully in 42 msec.
```

Query returned successfully in 42 m

Total rows: | Query complete 00:00:00.042 | CRLF | Ln 3

```
302
303
304
305  --4
306  RESET enable_indexscan;
307  RESET enable_bitmapscan;
308  RESET enable_indexonlyscan;
309
310  EXPLAIN ANALYZE
311  SELECT * FROM flights WHERE departing_airport_id = 10 AND arriving_airport_id = 25;
312
313
314
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	QUERY PLAN text
1	Index Scan using idx_flights_der_app on flights (cost=0.14..8.16 rows=1 width=84) (actual time=0.012..0.012 rows=0.00 loo…
2	Index Cond: ((departing_airport_id = 10) AND (arriving_airport_id = 25))
3	Index Searches: 1
4	Buffers: shared hit=1
5	Planning Time: 0.094 ms
6	Execution Time: 0.024 ms

✓ Successfully run. Total query runtime: 56 msec. 6 rows affecte

```
316  --5
317  EXPLAIN ANALYZE
318  SELECT * FROM flights WHERE departing_airport_id=10 AND arriving_airport_id=25;
319
320
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1

	QUERY PLAN text
1	Index Scan using idx_flights_der_app on flights (cost=0.14..8.16 rows=1 width=84) (actual time=0.012..0.012 rows=0.00 loo…
2	Index Cond: ((departing_airport_id = 10) AND (arriving_airport_id = 25))
3	Index Searches: 1
4	Buffers: shared hit=1
5	Planning Time: 0.093 ms
6	Execution Time: 0.027 ms

Successfully run. Total query runtime: 63 mse

```sql
--6
CREATE UNIQUE INDEX idx_passengers_passport_unique ON passengers(passport_number);


SELECT indexname, indexdef FROM pg_indexes
WHERE tablename = 'passengers' AND indexname = 'idx_passengers_passport_unique';


INSERT INTO passengers VALUES (200001, 'Test', 'One', '1990-05-05', 'Male', 'Kazakhstan', 'Kaz
INSERT INTO passengers VALUES (200002, 'Test', 'Two', '1995-07-07', 'Female', 'Kazakhstan', 'K
```

```sql
--7
CREATE INDEX idx_passengers_multi
ON passengers(first_name, last_name, date_of_birth, country_of_citizenship);

EXPLAIN ANALYZE
SELECT * FROM passengers
WHERE country_of_citizenship = 'Philippines'
  AND date_of_birth BETWEEN '1984-01-01' AND '1984-12-31';
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1

	QUERY PLAN text
1	Seq Scan on passengers (cost=0.00..6.50 rows=1 width=64) (actual time=0.025..0.025 rows=0.00 loops=1)
2	Filter: ((date_of_birth >= '1984-01-01'::date) AND (date_of_birth <= '1984-12-31'::date) AND ((country_of_citizenship)::text = 'Philippines':...
3	Rows Removed by Filter: 200
4	Buffers: shared hit=3
5	Planning:
6	Buffers: shared hit=42 read=1
7	Planning Time: 0.224 ms
8	Execution Time: 0.037 ms

Successfully run. Total query runtime: 64 msec

Total rows: 8 | Query complete 00:00:00.064

```sql
--8
SELECT indexname, indexdef FROM pg_indexes WHERE tablename = 'passengers';

DROP INDEX idx_passengers_multi;



SELECT indexname FROM pg_indexes WHERE tablename = 'passengers';
```

```sql
--9
EXPLAIN ANALYZE SELECT * FROM booking WHERE passenger_id = 100;

CREATE INDEX idx_booking_passenger ON booking(passenger_id);

EXPLAIN ANALYZE SELECT * FROM booking WHERE passenger_id = 100;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1

	QUERY PLAN text
1	Seq Scan on booking (cost=0.00..5.50 rows=4 width=53) (actual time=0.009..0.015 rows=4.00 loop...
2	Filter: (passenger_id = 100)
3	Rows Removed by Filter: 196
4	Buffers: shared hit=3
5	Planning:
6	Buffers: shared hit=5 read=1
7	Planning Time: 0.163 ms
8	Execution Time: 0.023 ms

Successfully run. Total query runtime: 69 ms

```sql
--10
CREATE INDEX idx_booking_flight_comp ON booking_flight(booking_id, flight_id);
CREATE INDEX idx_booking_price ON booking(ticket_price);
ANALYZE;

EXPLAIN ANALYZE
SELECT f.flight_no, b.ticket_price AS price
FROM booking b
JOIN booking_flight bf ON bf.booking_id = b.booking_id
JOIN flights f         ON f.flight_id   = bf.flight_id
WHERE b.ticket_price > 9000;
```

Data Output | Messages | Notifications

Showing rows: 1 to 21 | Page N

	QUERY PLAN text
1	Nested Loop (cost=5.78..10.60 rows=1 width=12) (actual time=0.099..0.105 rows=0.00 loops=1)
2	Buffers: shared hit=5
3	-> Hash Join (cost=5.51..10.05 rows=1 width=10) (actual time=0.099..0.104 rows=0.00 loops=1)
4	Hash Cond: (bf.booking_id = b.booking_id)
5	Buffers: shared hit=5
6	-> Seq Scan on booking_flight bf (cost=0.00..4.00 rows=200 width=8) (actual time=0.022..0.022 rows=1.00 loo...
7	Buffers: shared hit=2

```
375
376
377  --11
378  CREATE INDEX idx_flights_status ON flights(status);
379  ANALYZE flights;
380
381  EXPLAIN ANALYZE SELECT * FROM flights WHERE status = 'Departed';
382  EXPLAIN ANALYZE SELECT * FROM flights WHERE status = 'Boarding';
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No

	QUERY PLAN text
1	Index Scan using idx_flights_status on flights (cost=0.14..6.70 rows=1 width=84) (actual time=0.019..0.019 rows=0.00 loo…
2	Index Cond: ((status)::text = 'Boarding'::text)
3	Index Searches: 1
4	Buffers: shared read=1
5	Planning Time: 0.028 ms
6	Execution Time: 0.024 ms

```
384
385
386  --12
387  CREATE INDEX idx_baggage_weight ON baggage(weight_in_kg);
388  ANALYZE baggage;
389
390  EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg > 49;
391  EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg > 5;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1

	QUERY PLAN text
1	Seq Scan on baggage (cost=0.00..6250.00 rows=299935 width=30) (actual time=0.019..69.813 rows=299961.00 loo…
2	Filter: (weight_in_kg > '5'::numeric)
3	Rows Removed by Filter: 39
4	Buffers: shared hit=2500
5	Planning:
6	Buffers: shared hit=2 read=2
7	Planning Time: 0.244 ms
8	Execution Time: 91.968 ms

Successfully run. Total query runtime

Total rows: 8 | Query complete 00:00:00.910

```
394
395  --13
396  CREATE INDEX idx_airport_name ON airport(airport_name);
397  CREATE INDEX idx_airport_name_pattern ON airport(airport_name text_pattern_ops);
398  ANALYZE airport;
399
400  EXPLAIN ANALYZE SELECT * FROM airport WHERE airport_name LIKE 'A%';
401  EXPLAIN ANALYZE SELECT * FROM airport WHERE airport_name LIKE '%Airport';
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on airport (cost=0.00..2487.00 rows=99990 width=66) (actual time=0.021..26.960 rows=100000.00 loo...
2	Filter: ((airport_name)::text ~~ '%Airport'::text)
3	Buffers: shared hit=1237
4	Planning Time: 0.146 ms
5	Execution Time: 34.077 ms

✓ Successfully run. Total query runtime: 1 secs 883 msec. 5 row

```
403
404
405  --14
406  CREATE INDEX idx_airport_country_city ON airport(country, city);
407  ANALYZE airport;
408
409  EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Country 7';
410  EXPLAIN ANALYZE SELECT * FROM airport WHERE city = 'City 150';
411  EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Country 7' AND city = 'City 150';
```

Wrappers

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No:

	QUERY PLAN text
1	Index Scan using idx_airport_country_city on airport (cost=0.42..8.44 rows=1 width=66) (actual time=0.029..0.030 rows=1.00 loo...
2	Index Cond: (((country)::text = 'Country 7'::text) AND ((city)::text = 'City 150'::text))
3	Index Searches: 1
4	Buffers: shared hit=4
5	Planning Time: 0.172 ms
6	Execution Time: 0.051 ms

Successfully run. Total query runtime: 2 s

Total rows: 6 | Query complete 00:00:02.487

```
413
414
415  --15
416  CREATE INDEX idx_passengers_lastname_lower ON passengers(LOWER(last_name));
417  ANALYZE passengers;
418
419  EXPLAIN ANALYZE SELECT * FROM passengers WHERE LOWER(last_name) = 'irnis';
420  EXPLAIN ANALYZE SELECT * FROM passengers WHERE last_name = 'Irnis';
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on passengers (cost=0.00..5.50 rows=1 width=64) (actual time=0.011..0.011 rows=0.00 loop...
2	Filter: ((last_name)::text = 'Irnis'::text)
3	Rows Removed by Filter: 200
4	Buffers: shared hit=3
5	Planning Time: 0.018 ms
6	Execution Time: 0.014 ms

Successfully run. Total query runtime: 55 msec. 6 rows affect

```
422
423  --16
424  CREATE INDEX idx_booking_pass_price ON booking(passenger_id, ticket_price);
425  VACUUM ANALYZE booking;   -- важно для Index Only Scan
426
427  EXPLAIN ANALYZE SELECT ticket_price FROM booking WHERE passenger_id = 100;
428
```

```
--17
INSERT INTO airline VALUES
 (201, 'AL001', 'Test Air 1', 'Kazakhstan', now(), now()),
 (202, 'AL001', 'Test Air 2', 'Turkey', now(), now());

CREATE UNIQUE INDEX idx_airline_code_unique ON airline(airline_code);
SELECT airline_code, COUNT(*) FROM airline GROUP BY airline_code HAVING COUNT(*) > 1;
SELECT * FROM airline WHERE airline_code = 'AL001';

UPDATE airline SET airline_code = 'AL' || lpad(airline_id::text, 3, '0') WHERE airline_id IN (

CREATE UNIQUE INDEX idx_airline_code_unique ON airline(airline_code);
```